HealthBridge Multi-Member Architecture

This document explains the current multi-member GPS telemetry architecture used in the HealthBridge Android application.

1. Publisher Concept

Each Android device publishes GPS telemetry for exactly ONE member. The member identity is defined by the TelemetryEngine configuration.

```
telemetryEngine =
    TelemetryEngine(this, "alain")
```

Meaning:

"This phone belongs to Alain and publishes Alain's GPS coordinates."

2. Viewer Concept

Any device can listen to one or more members and display them on the map.

```
listenToMember("alain")
listenToMember("mary")
```

Meaning:

"Display Alain and Mary on the map."

3. Owner of Phone vs Other Members

The phone owner is the member whose telemetry is published by the device. Other members are only displayed on the map through Firebase listeners.

Concept	Meaning
TelemetryEngine(this, "alain")	This device publishes Alain GPS data
listenToMember("mary")	Show Mary on the map
listenToMember("alain")	Show Alain on the map

4. Example Family Setup

Example:

Samsung Tablet:
TelemetryEngine(this, "alain")

Motorola Phone:
TelemetryEngine(this, "mary")

Both devices can listen to:
listenToMember("alain")
listenToMember("mary")

Result: Both devices show two live moving markers.

The HealthBridge architecture now supports true multi-member live telemetry, where each device can simultaneously act as:

• a GPS publisher
• a Firebase telemetry source
• a multi-member viewer
• a caregiver monitoring device